High-Frequency Trading (HFT)

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Outline

The rise of HFT I

Floor and phone traders used to dominate exchanges.

But that was in the past . . .

Source: http://www.benkepple.com

The rise of HFT II

Three big evolutions in financial markets

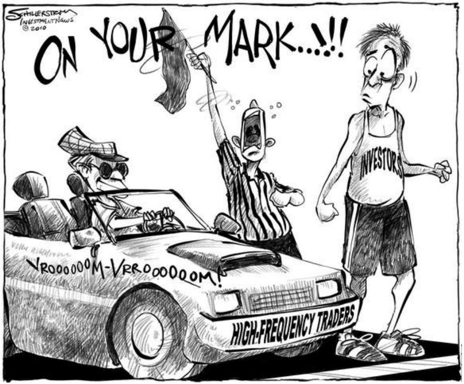
1. Automatisation: markets became electronic.
2. Fragmentation: stock markets compete with alternative venues and dark pools.
3. Information: faster, richer and digital real-time data.

The birth of algorithmic trading

- Systematic investment decisions: trading rules are coded, prices are monitored by dedicated programs.
- Productivity gains via automatisation: less human resources, less repetitive tasks.
- Widespread use of Quantitative Finance tools: portfolio optimisation, pricing, execution protocols, etc.

$\Rightarrow$ Algorithmic trading: traders monitor whilst robots execute.

The rise of HFT III



HFTs = algo traders + high speed.
Source: http://www.investmentnews.com

The rise of HFT IV

The birth of HFT

- Algorithmic trading: robot traders are faster, more reactive and more accurate than human traders.
- Information accesibility at very high speed: real-time data flow with low latency e.g. direct market access and co-location.
- Technological advances in computing: data processing, parallel computing, over-clocking, etc.

$\Rightarrow$ HFT = algo traders using speed as their main advantage.

- **Remark**: HFT does not necessarily mean very frequent trading.

The rise of HFT V

Exchanges now are just servers.
HFTs have their servers there as well: co-location.
Source: http://www.wired.co.uk

HFTs are heterogeneous

HFT strategies: The SEC definition

1. Market-making.
2. HF arbitrage.
3. HF directional.
4. Manipulators.

These strategies are not new: the novelty is the use of speed as a competitive advantage.

Alternative definition of HFTs based on liquidity + strategy

1. Makers: market-makers.
2. Takers: arbitrage and directional.
3. Gamers.

HF Makers I

Definition

- They are liquidity providers, i.e. they use LOs.
- They play the role of dealers: they offer ask and bid quotes, earning the spread.
- For liquid stocks, spread $= 0.01$ USD:
  - ⋆ It is a very small potential gain per trade.
  - ⋆ But if there are lots of trades, the gain can be important.
- Speed is important:
  - ⋆ Fast and frequent trades: earn the spread as many times as possible.
  - ⋆ Priority in the LOB: be the first in price and time to enhance execution.
  - ⋆ High reactivity: quick response to liquidity and market fluctuations.

HF Makers II

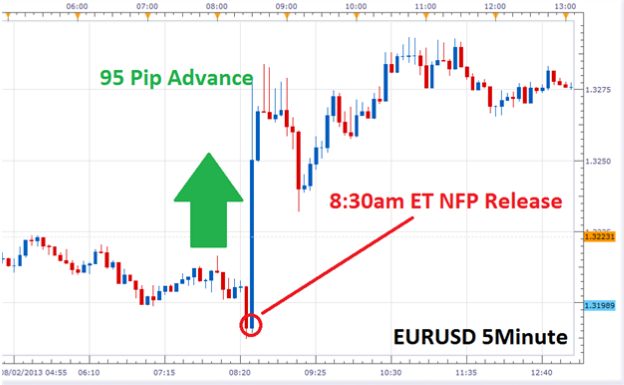
HF market-makers: faster, more reactive, more resilient.
Source: http://williambanzai7.blogspot.mx

HF Takers I

Definition

- They are liquidity consumers, i.e. they use MOs.
- They are not dealers: their gain does not come from the spread.
- They use speed to capture profitable opportunities before others:
  - ⋆ Arbitrage e.g. correlations and cross-market.
  - ⋆ Directional strategies e.g. news trading.
- In general, HF takers consume liquidity from HF makers:
  - ⋆ HF makers and takers play between them: zero-sum game.
  - ⋆ They do not make their profits out of retail traders.

HF Takers II



HF directional: in news trading, faster means richer. 95 pips $\sim$ 0.7%.
Source: http://www.dailyfx.com

HF Gamers I

Definition

- They exploit structural deficiencies of electronic markets.
- They use both LOs and MOs, according to their strategy.

Examples of HF gamer strategies

- Spoofing: you place sell LOs not meant to be executed, giving a false impression of selling shares, when your real order is a buy.
- Momentum ignition: you lure traders to trade quickly and cause a rapid price move. This is done by trading highly correlated instruments.
- Flash trades: you can "see" orders arriving to the LOB before they are made public, and make a profit by front-running them.
- Stuffing: you send and immediately cancel lots of LOs to confuse markets and traders.

HF Gamers II

HF gaming is like Poker: force other players to errors and be the first to seize opportunities, but it should remain within the limits of what is accepted.
Source: http://www.zerochan.net

HFT vs financial efficiency and stability I

HFTs are mutable

- Makers can become takers when they need to reduce their inventory.
- Takers can become makers when they set a take-profit order.
- Gamers can behave as makers or takers since they use both MOs and LOs to exploit infrastructure cracks.

But HFTs are specialised in strategies

- Most of the time:
  - ⋆ Makers use LOs on both ask and bid sides.
  - ⋆ Takers use MOs and one-sided LOs.
  - ⋆ Gamers exploit market "glitches".

$\Rightarrow$ **Differentiate between HFT strategies** to assess their systemic impact and risk.

HFT vs financial efficiency and stability II

Gamers can be harmful

There is some consensus on "bad" HTF strategies:

- Gamers manipulate market prices with artificial quotes: spoofing.
- They can cause short-term market disruptions: momentum ignition.
- They have "insider trader" behaviour: flash trades.
- They overflow LOBs with noise orders that are immediately cancelled: stuffing.

But makers and takers are rather beneficial

There is some consensus on "good" HTF strategies:

- HF makers improve liquidity and reduce spreads.
- HF takers (arbitrageurs) instantaneously correct market inefficiencies.
- Concerning HF makers and takers vs market volatility:
  - ⋆ There is no consensus in empirical studies.
  - ⋆ But it seems HT makers and takers in general do not increase it.
  - ⋆ And in some cases they even decrease it.

HFT vs financial efficiency and stability III

Don't blame gamers, blame markets and regulators

Gamers only use what markets and regulators allow them to:

- The National Best Bid and Offer (NBBO) and zero margin calls allow spoofing.
- The 5ms latency between NY (NYSE) and Chicago (CME) allows for momentum ignition.
- Flash trades are legal in several markets.
- If electronic markets get stuffed with noise orders is because they need better infrastructure and/or better LOB rules.

Market impact I

Statistical properties of MOs

- The average size of a buy MO is the volume at the best ask:
  - ⋆ In general, MOs only consume one level of the LOB.
- There is hidden liquidity in the LOB:
  - ⋆ If the best ask is consumed, it can be instantly replenished with icebergs.
- There is latent liquidity in the market:
  - ⋆ When there are many buy MOs hitting the ask, new liquidity providers can appear with LOs at the best ask.
- There is resilience and adaptability in the market:
  - ⋆ When there is a predictable buying pattern of MOs (e.g. execution algo), the market is less reactive to buy MOs than to sell MOs.

Market impact II

Why market impact is concave

Let v be the size of the MO and $h(v)$ the market-impact function:

- As v increases, the MO consumes more liquidity and tests market's depth.

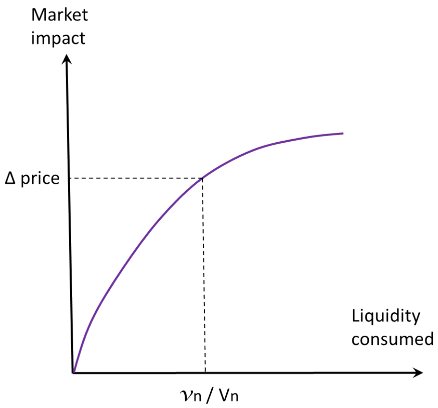
$\Rightarrow$ $v \mapsto h(v)$ is increasing, i.e. $h'(v) > 0$.

- But as we saw, market's resilience and adaptability increases in v.

$\Rightarrow$ $v \mapsto h'(v)$ is decreasing, i.e. $h''(v) < 0$.

- Therefore, the market impact $h(v)$ is concave in the size v of the MO.
- Recall that in Kyle model, the market impact was assumed linear.

Market impact III



Market impact is concave.

Optimal trading curves I

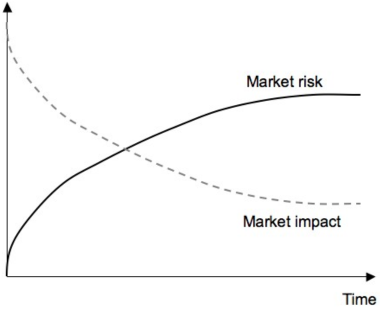
Trader's dilemma

- If we trade slow, prices will move away from their current quote.

$\Rightarrow$ Market risk.

- If we trade fast, our order will drive prices away from the current quote.

$\Rightarrow$ Market (or price) impact.



Optimal trading curves II

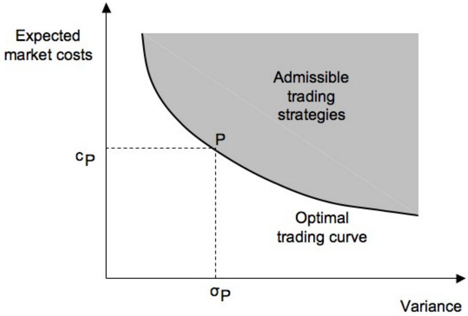
Optimal trading curve

- In the Markowitz portfolio we minimise the risk whilst maximising the return.

$\Rightarrow$ Efficient frontier.

- Following this idea, we can minimise both market impact and market risk.

$\Rightarrow$ Optimal trading curve.



Almgren-Chriss: model I

Execution times and trade sizes

- Assume we decided to execute N trades at evenly-distributed times:

$$0 = t_0 < t_1 < t_2 < \cdots < t_N = T\,, \quad t_n - t_{n-1} = \tau \quad \text{constant} \quad \forall n\,.$$

- Every time t_n we buy v_n shares. This defines the trading curve

$$(v_1, \ldots, v_N)\,, \qquad \sum_{n=1}^{N} v_n = v^*$$

- The goal is to find the optimal trading curve $(v_1, \ldots, v_N)$.

Almgren-Chriss: model II

Market impact function

Based on Almgren 2001, Almgren *et al* 2005 and Bouchaud 2003 we define :

$$h(v_n) = \kappa \sigma_n \tau^{1/2} \left(\frac{v_n}{V_n} \right)^{\gamma}$$

where

- v_n is the number of shares we traded at time t_n.
- σ_n and V_n are the intraday volatility and volume curves, respectively.
- $\kappa > 0$ and $\gamma \in (0, 1)$ are the market-impact parameters.

Empirically $\gamma \sim 1/2$, but it can be calibrated individually for each stock.

Almgren-Chriss: model III

Price model

Assume a Brownian motion model:

$$S_{n+1} = S_n + \sigma_{n+1}\tau^{1/2}\varepsilon_{n+1}\,, \qquad \varepsilon_n \sim \mathcal{N}(0,1) \quad \text{i.i.d}\,.$$

Any martingale can be used, provided $(\varepsilon_n)_{n=1}^N$ are i.i.d. of mean zero and variance 1.

Wealth process

$$\begin{aligned} W(v_1,\ldots,v_N) &= \sum_{n=1}^{N} v_n(S_n + h(v_n)) \\ &= \sum_{n=1}^{N} v_n S_n + \sum_{n=1}^{N} \kappa\sigma_n\tau^{1/2} v_n \left(\frac{v_n}{V_n}\right)^{\gamma} \\ &= \text{ideal cost} + \text{market impact} \end{aligned}$$

Almgren-Chriss: solution I

Implementation Shortfall (IS)

- For an IS algorithm, the benchmark is the price at the moment when the execution starts.

$\Rightarrow$ The relative wealth process is thus

$$W^{\sharp}(v_1, \ldots, v_N) = W - S_0 \sum_{n=1}^{N} v_n \, .$$

Change of variables

$$x_n := \sum_{i=n}^{N} v_i \qquad \Longleftrightarrow \qquad v_n = x_n - x_{n+1}$$

Almgren-Chriss: solution II

Relative wealth process for IS

After **some algebra**, it can be shown that

$$\begin{aligned} W^{\sharp}(x_1,\ldots,x_N) &= \sum_{n=1}^{N} x_n \sigma_n \tau^{1/2} \varepsilon_n + \sum_{n=1}^{N} \kappa \sigma_n \tau^{1/2} \frac{(x_n - x_{n-1})^{\gamma+1}}{V_n^{\gamma}} \\ &= \left(\sum_{n=1}^{N} x_n \sigma_n \varepsilon_n + \sum_{n=1}^{N} \kappa \sigma_n \frac{(x_n - x_{n-1})^{\gamma+1}}{V_n^{\gamma}} \right) \tau^{1/2} . \end{aligned}$$

Normalised relative wealth

We will consider the relative wealth per time unit, i.e.

$$\begin{aligned} \tilde{W}(x_1,\ldots,x_N) &:= \frac{W^{\sharp}}{\tau^{1/2}} \\ &= \sum_{n=1}^{N} x_n \sigma_n \varepsilon_n + \sum_{n=1}^{N} \kappa \sigma_n \frac{(x_n - x_{n-1})^{\gamma+1}}{V_n^{\gamma}} \end{aligned}$$

Almgren-Chriss: solution III

Mean and variance

$$\mathbb{E}(\tilde{W}) = \sum_{n=1}^{N} \kappa \sigma_n \frac{(x_n - x_{n+1})^{\gamma+1}}{V_n^{\gamma}}, \qquad \mathbb{V}(\tilde{W}) = \sum_{n=1}^{N} x_n^2 \sigma_n^2$$

Cost functional

$$\begin{aligned} J_\lambda(x_1, \ldots, x_N) &= \mathbb{E}(\tilde{W}) + \lambda \mathbb{V}(\tilde{W}) \\ &= \sum_{n=1}^{N} \kappa \sigma_n \frac{(x_n - x_{n+1})^{\gamma+1}}{V_n^{\gamma}} + \lambda \sum_{n=1}^{N} x_n^2 \sigma_n^2 \end{aligned}$$

where $\lambda > 0$ is the risk-aversion parameter. Observe that

$$J_\lambda(x_1, \ldots, x_N) = \text{market impact} + \lambda \times \text{market risk}$$

Almgren-Chriss: solution IV

Optimality condition

The critical points of J_λ are found by solving $\partial J_\lambda / \partial x_n = 0$ for all n:

$$\kappa\sigma_n(\gamma+1)\frac{(x_n - x_{n+1})^\gamma}{V_n^\gamma} - \kappa\sigma_{n-1}(\gamma+1)\frac{(x_{n-1} - x_n)^\gamma}{V_{n+1}^\gamma} + 2\lambda\sigma_n^2 x_n = 0\,.$$

Optimal trading curve

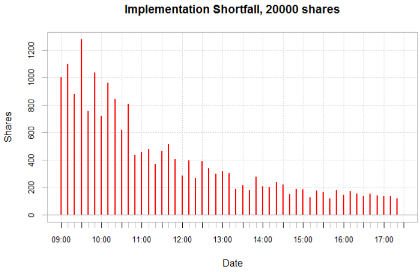
The optimal trading curve $(v_1, \ldots, v_N)$ for the IS algo is then

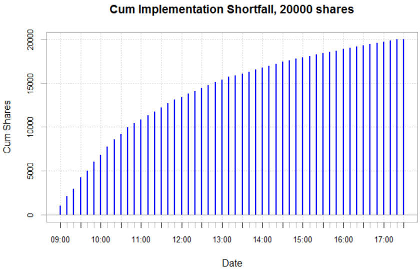
$$v_{n-1} = V_{n-1}\left[\frac{\sigma_n}{\sigma_{n-1}}\left(\frac{v_n}{V_n}\right)^\gamma + \frac{2\lambda}{\kappa(\gamma+1)}\frac{\sigma_n^2}{\sigma_{n-1}}\left(\sum_{i=n}^N v_i\right)\right]^{1/\gamma}$$

with the conditions

$$v_0 = 0\,, \qquad v_{N+1} = 0\,, \qquad \sum_{n=1}^N v_n = v^*$$

Almgren-Chriss: numerical simulations I



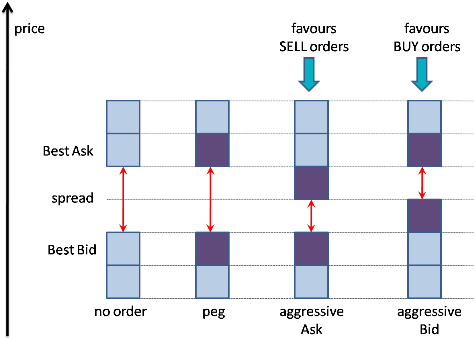


Numerical example of the IS algorithm.

Market-making: rules of the game I

What is a market-maker (MM)?

- A trader who posts firm buying (bid) and selling (ask) quotes on the LOB.
- Liquidity provider $\Rightarrow$ earns the spread.



Market-making: rules of the game II

Risks for a MM

- Adverse selection: If a MM sells an asset it is not necessarily good news.
- Inventory risk: Uncertainty on the execution of her limit orders.
- Mean-reversion strategy: MMs sell when assets go up, buy when assets go down.

$\Rightarrow$ Potential risks on trends.

Strategy of a MM

- MMs use the spread to control inventory and compensate from adverse selection.
- MMs lose money vs informed traders but make money vs noise traders.

Stochastic control: state variables

State variables in a Markovian world

- The mid-price $S(t)$, e.g. a jump process or an Itô diffusion.
- The half market spread $Z(t)$:
  - Best ask $= S(t) + Z(t)$, best bid $= S(t) - Z(t)$
- The volatility $\Sigma(t)$.
- The market-maker's quotes $p^{\pm}$ and her controls $\delta^{\pm}$:

$$p^{+}(t) = S(t) + \delta^{+}\,, \quad p^{-}(t) = S(t) - \delta^{-}\,.$$

- The inventory $Q(t)$:

$$dQ(t) = dN^{-}(t) - dN^{+}(t)\,,$$

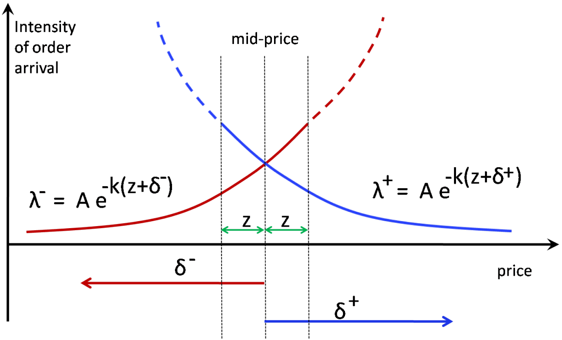
  where $dN^{+}(t)$ and $dN^{-}(t)$ are two independent Poisson processes of intensity

$$\lambda^{\pm}(\delta^{\pm}) = Ae^{-K(t)[z+\delta^{\pm}]}\,.$$

- The cash $X(t)$:

$$dX(t) = [S(t) + \delta^{+}]dN^{+}(t) - [S(t) - \delta^{-}]dN^{-}(t)\,.$$

Stochastic control: arrival of MOs



Market order intensities $\lambda^{\pm}$ are extrapolated when $\delta^{\pm} \leq -z$ (dotted lines).

Stochastic control: HJB equation

Controls

- From all state variables, the MM can only control δ^+ and δ^-.
- We will denote $Y(t)$ the (Markovian) vector of non-controlled variables:

$$Y(t) = (S(t), \Sigma(t), Z(t), \dots)$$

Value function when utility = PNL

$$u(t,y,q,x) = \max_{\delta^\pm \in \mathcal{A}} \mathbb{E}_{t,y,q,x}[X(T) + Q(T)S(T)]\,.$$

Hamilton-Jacobi-Bellman (HJB) equation

$$\begin{aligned}(\partial_t + \mathcal{L})\,u + \max_{\delta^+ \in \mathcal{A}} Ae^{-k[z+\delta^+]}\left[u(t,y,q-1,x+(s+\delta^+)) - u(t,s,q,x)\right] \\ + \max_{\delta^- \in \mathcal{A}} Ae^{-k[z+\delta^-]}\left[u(t,y,q+1,x-(s-\delta^-)) - u(t,y,q,x)\right] = 0 \\ u(T,y,q,x) = x + qs\end{aligned}$$

Stochastic control: inventory penalties and transaction costs I

Inventory penalties

- A penalty at expiry, depending on the spread:

$$\Pi_1(T) = \eta Z(T) Q^2(T), \quad \eta \geq 0.$$

$\Rightarrow$ Transaction costs for clearing inventory at $t = T$ with a market order.

- An integral (path-dependent) penalty, depending on the volatility:

$$\Pi_2(T) = \nu \int_t^T \Sigma^2(\xi) Q^2(\xi)\, d\xi, \quad \nu \geq 0.$$

$\Rightarrow$ Tracking error with respect to a flat-inventory position.

Stochastic control: inventory penalties and transaction costs II

Value function when utility = PNL − inventory penalty

$$u(t,y,q,x) = \max_{\delta^{\pm}\in\mathcal{A}} \mathbb{E}_{t,y,q,x}\Big[X(T)+Q(T)S(T)-\varepsilon\Pi(T)\Big]\,, \quad \Pi := \Pi_1+\Pi_2\,.$$

HJB with inventory penalty and transaction costs

$$(\partial_t+\mathcal{L})\,u + \max_{\delta^+\in\mathcal{A}} Ae^{-k[z+\delta^+]}\left[u(t,y,q-1,x+(s+\delta^+)-\alpha)-u(t,y,q,x)\right]$$

$$+\max_{\delta^-\in\mathcal{A}} Ae^{-k[z+\delta^-]}\left[u(t,y,q+1,x-(s-\delta^-)-\alpha)-u(t,y,q,x)\right] = \varepsilon\nu\sigma^2 q^2$$

$$u(T,y,q,x) = x+sq-\varepsilon\eta z q^2$$

Stochastic controls: solution

Optimal controls

$$\begin{aligned}
\psi_{\alpha *} &= \frac{2}{k} + 2\alpha + 2\varepsilon\tilde{\pi} + O\left(\varepsilon^2\right) \quad \text{(MM's spread)} \\
r_{\alpha *} &= s + \Delta - 2\varepsilon q \tilde{\pi} + O\left(\varepsilon^2\right) \quad \text{(centre of the MM's spread)}
\end{aligned}$$

where

$$\begin{aligned}
\Delta &:= \mathbb{E}_{t,y}\left[S(T)\right] - s \qquad \text{(directional bet)} \\
\tilde{\pi} &:= \eta \mathbb{E}_{t,y}\left[Z(T)\right] + \nu \mathbb{E}_{t,y}\left[\int_t^T \Sigma^2 \, d\xi\right] \\
&= \eta \times \text{expected spread} + \nu \times \text{expected volatility}
\end{aligned}$$

Stochastic controls: remarks I

Expected gains per trade

- The expected gain per traded spread is $\psi_{\alpha*} = \psi_{0*} + 2\alpha$.
- The MM pays 2α per traded spread.

$\Rightarrow$ The expected gain per traded spread is constant and equal to ψ_{0*}.

Inventory management

- $q > 0$ and $\Delta = 0$ and $\Rightarrow r_{\alpha*} < s$, i.e. the MM is rather selling.
- $q < 0$ and $\Delta = 0$ and $\Rightarrow r_{\alpha*} > s$, i.e. the MM is rather buying.

Directional bet

- $\Delta > 0$ and $q = 0 \Rightarrow r_{\alpha*} > s$, i.e. the MM is rather buying.
- $\Delta < 0$ and $q = 0 \Rightarrow r_{\alpha*} < s$, i.e. the MM is rather selling.

Stochastic controls: remarks II

The effect of transaction costs

If $\alpha > 0$:

- The MM compensates their loss in transaction costs by widening the spread.

$\Rightarrow$ Gain per traded spread constant but smaller probability of execution.

- If all MMs have wider spreads $\Rightarrow$ bigger market spreads, hence less liquidity.

If $\alpha < 0$ i.e. there is a rebate:

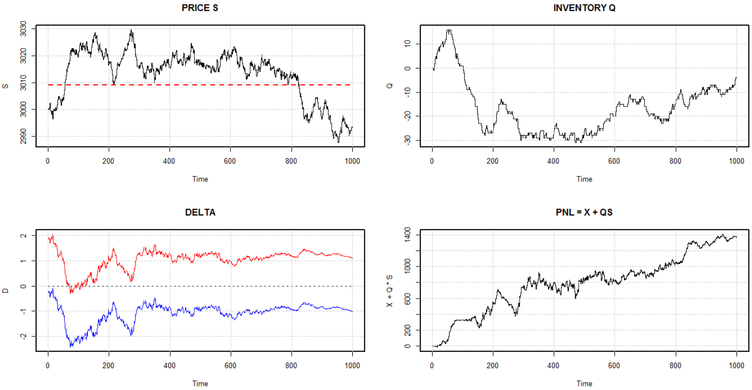
- The MM systematically reduces their spread.

$\Rightarrow$ Gain per traded spread constant but bigger probability of execution.

- The MM could even buy and sell at the same price, earning no profit except for the rebate.

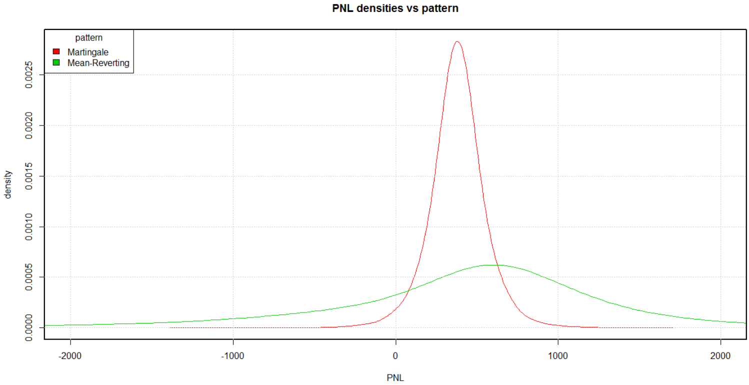
$\Rightarrow$ Scalping or rebate arbitrage.

Numerical simulations: typical trading day with mean-reversion



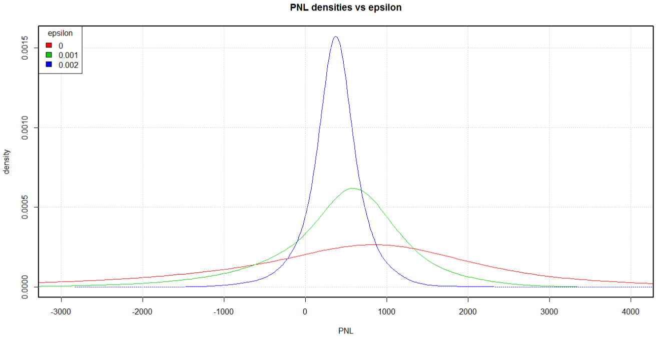
$n = 1000$, $s = 3000$, $\mu = 3009$ $(+0.3\%)$, $\varepsilon = 0.001$, $z = 0.5$, $\alpha = 0.05$.

Numerical simulations: martingale vs mean-reversion



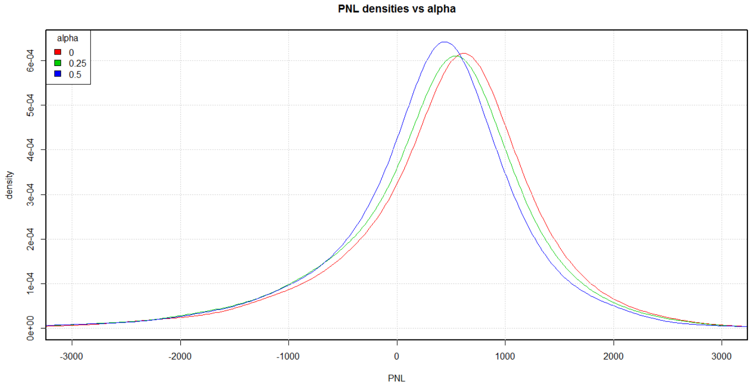
Simulations $= 10k$, $n = 1000$, $s = 3000$, $\mu = 3009$ $(+0.3\%)$, $z = 0.5$, $\varepsilon = 0.001$, $\alpha = 0.05$.

Numerical simulations: effect of inventory risk ε



Simulations $= 10k$, $n = 1000$, $s = 3000$, $\mu = 3009$ $(+0.3\%)$, $z = 0.5$, $\alpha = 0.05$.

Numerical simulations: effect of transaction costs α



Simulations $= 10k$, $n = 1000$, $s = 3000$, $\mu = 3009$ $(+0.3\%)$, $z = 0.5$, $\varepsilon = 0.001$.

Final comments

Summary of this presentation

- We explained what is HFT and reviewed the different kinds of HFT players.
- We saw in detail the effect of market impact.
- We worked an example of optimal execution (IS) via mean-variance analysis.
- We worked an example of optimal HF market-making with stochastic control.
- We saw the risk profile and PNL distribution of a market-maker, and how it varies on inventory aversion and transaction costs.

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